

GymX: User Classes & Functional Requirements

This document presents **Part 2** of the GymX Software Requirements Specification (SRS), detailing the complete user role hierarchy and the full set of functional modules that define the platform's capabilities. GymX is a comprehensive gym management system designed to serve every stakeholder – from super administrators to registered members – through a robust Role-Based Access Control (RBAC) model and a well-structured suite of 15 functional requirements.

The document is organised into two major sections. **Section 14** defines eleven distinct user classes, each with clearly delineated responsibilities and permissions. **Section 15** enumerates the functional requirements across fifteen modules, covering everything from authentication and user management to AI-powered recommendations. This specification serves as the authoritative reference for product managers, developers, and QA engineers building and validating the GymX platform.

SRS PART 2

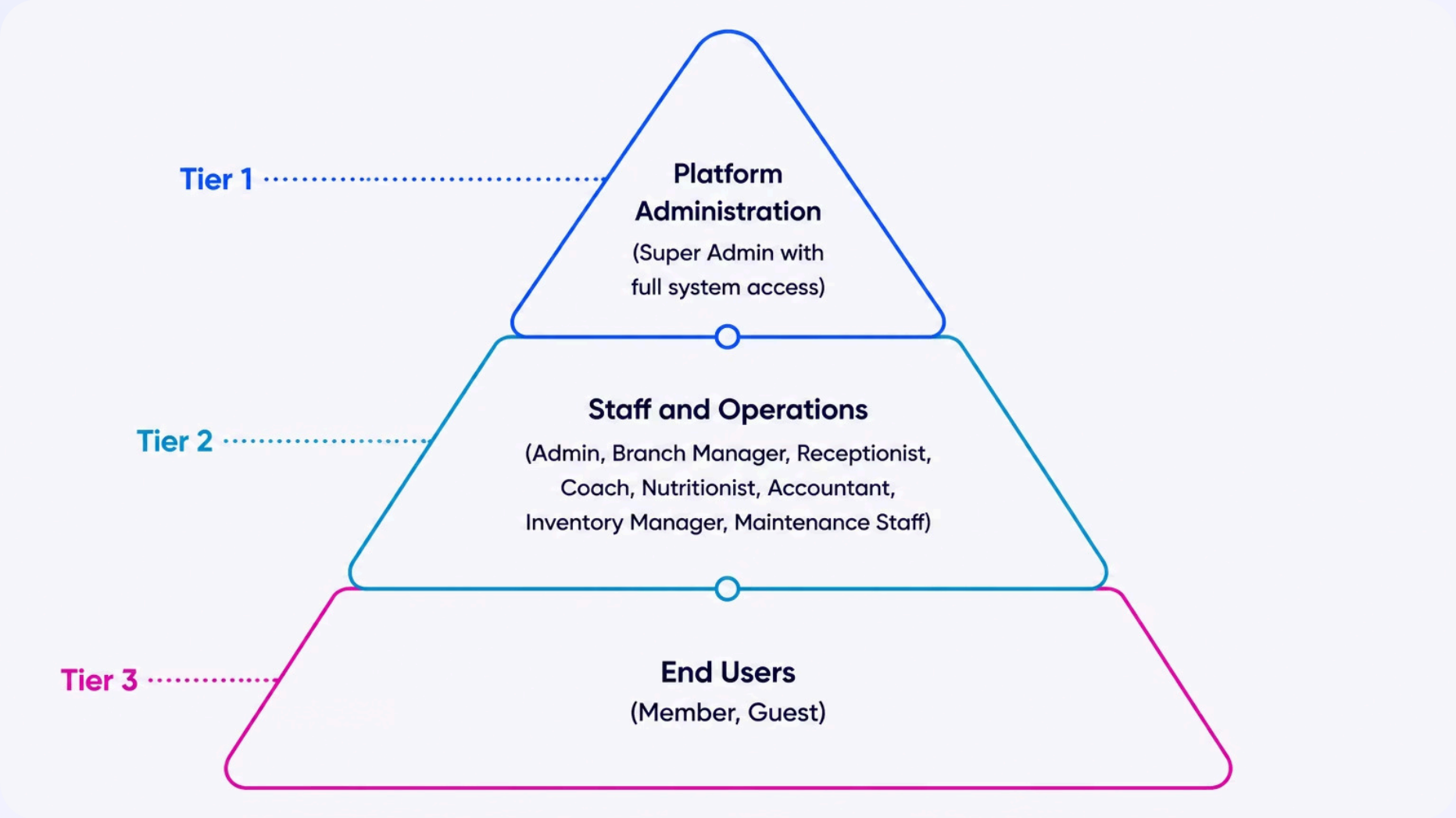
11 USER CLASSES

15 FUNCTIONAL MODULES

RBAC MODEL

Role-Based Access Control: Overview

GymX is built on a **Role-Based Access Control (RBAC)** model, meaning every user is assigned a specific role that determines their permissions, accessible features, and scope of responsibility within the platform. This architecture ensures security, operational clarity, and a tailored experience for each user type. The eleven roles are organised into three tiers: **Platform Administration**, **Staff & Operations**, and **End Users**.



The pyramid above illustrates the three-tier hierarchy. At the apex sits the **Super Admin** with unrestricted platform-wide access. The middle tier encompasses eight staff and operations roles, each scoped to specific functional domains or branch-level responsibilities. The base tier represents the end users – Members and Guests – who interact with the platform primarily for personal fitness management and information access.

Platform Administration Roles

The top tier of the GymX RBAC model comprises two roles with the broadest system access: the **Super Admin** and the **Admin**. These roles are responsible for platform configuration, user governance, and the overall health of the system. They operate at different scopes – the Super Admin across the entire platform, and the Admin within a specific gym branch.

14.1 Super Admin

The Super Admin holds **full access** to the entire GymX platform. This role is responsible for global system configuration, multi-branch management, and platform-level security. The Super Admin is the ultimate authority within the system and can perform any action across all modules.

- Manage all branches across the platform
- Create and manage administrators
- Configure global system settings
- View all reports across branches
- Monitor system-wide activity
- Manage roles and permissions
- Backup and restore the system

14.2 Admin

The Admin manages the **daily operations** of a specific gym or branch. While their scope is limited to a single branch, they have comprehensive control over members, employees, memberships, payments, and classes within that branch. Admins serve as the primary operational decision-makers at branch level.

- Manage members and memberships
- Manage employees and staff assignments
- Approve payments and transactions
- View branch-level reports
- Manage fitness class schedules



The Admin role is scoped to a single branch, unlike the Super Admin who has platform-wide access.

Staff & Operations Roles

The middle tier of the RBAC model comprises eight staff roles, each responsible for a specific operational domain within the gym. These roles ensure that every aspect of gym operations – from front desk management to equipment maintenance – has a dedicated owner with the appropriate system permissions. Each role is designed to minimise overlap while maximising accountability.



14.3 Branch Manager

Oversees a single gym branch, monitoring employees, managing attendance, supervising daily operations, reviewing branch performance, and generating operational reports.



14.4 Receptionist

Handles front desk operations including registering new members, renewing memberships, recording attendance, scheduling appointments, and handling customer inquiries.



14.5 Coach

Creates workout plans, tracks member progress, schedules training sessions, updates exercise routines, and evaluates individual member performance over time.



14.6 Nutritionist

Creates diet plans, monitors nutrition progress, updates meal plans, and calculates calorie requirements tailored to each member's fitness goals.



14.7 Accountant

Manages invoices, records payments, generates financial reports, tracks expenses, and oversees revenue management for the branch or platform.



14.8 Inventory Manager

Adds products, updates stock levels, records purchases, and manages supplier relationships to ensure the gym's inventory is always up to date.



14.9 Maintenance Staff

Schedules maintenance tasks, reports damaged equipment, and updates maintenance history to keep all gym equipment in safe, working condition.

End User Roles

The base tier of the GymX RBAC model consists of two end user roles: the **Member** and the **Guest**. These roles represent the primary consumers of the platform's services. Members enjoy a rich set of self-service capabilities, while Guests have limited access designed to facilitate onboarding and conversion to registered membership.

14.10 Member

A **registered gym customer** with full access to their personal dashboard and self-service features. Members can manage their fitness journey, handle payments, and interact with coaches and nutritionists through the platform.

- View membership details and history
- Access personalised workout plan
- Access personalised diet plan
- Book fitness classes online
- View personal attendance records
- Update profile information
- Make payments securely

14.11 Guest

An **unregistered visitor** with minimal platform access. The Guest role is designed to provide just enough information and functionality to encourage registration and conversion to a paying Member. Guests cannot access personal fitness features or make bookings.

- Browse gym information and facilities
- Register for a new account
- Contact support or sales team



Guests are a key conversion point. The registration flow should be frictionless to maximise Member acquisition.

Functional Requirements: Authentication & User Management

FR-001

FR-002

The first two functional modules form the foundation of the GymX platform's security and user governance. **FR-001** ensures that all users can authenticate securely, while **FR-002** gives administrators the tools to manage the entire user lifecycle. Together, these modules enforce the RBAC model and protect the platform from unauthorised access.

FR-001: Authentication Module

Provides **secure authentication** for all users across all roles. This module is the first line of defence for the platform and must support both current requirements and future enhancements such as Multi-Factor Authentication.

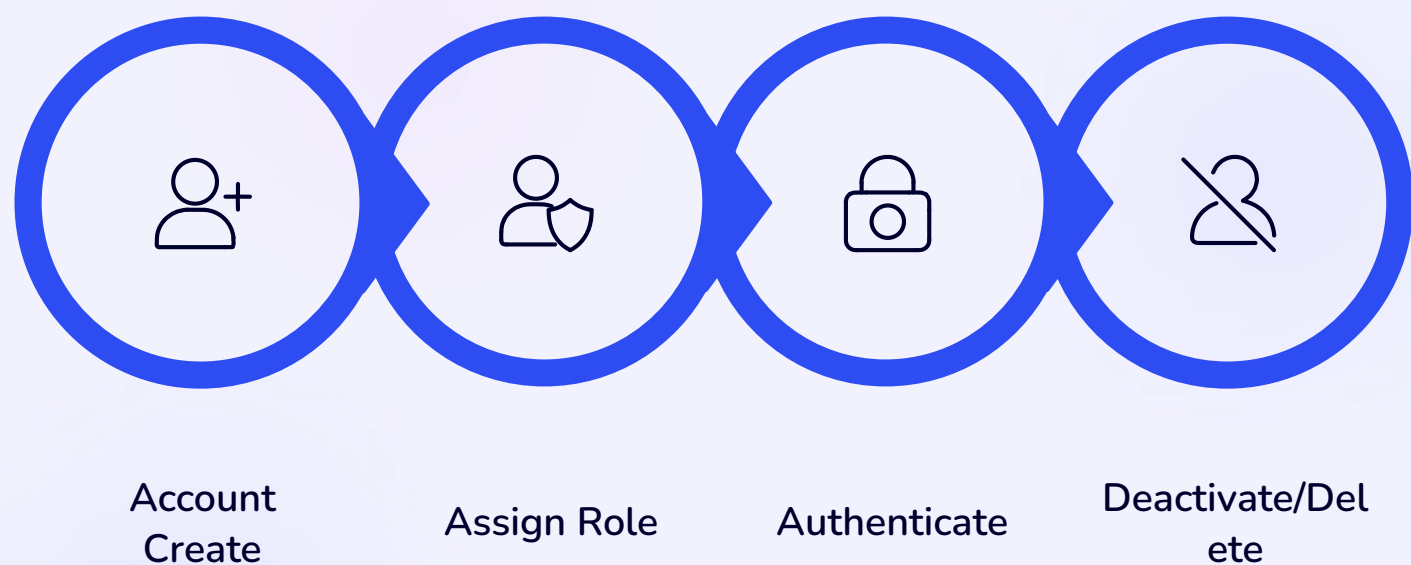
- Login and Logout
- Forgot Password and Reset Password
- Change Password
- Email Verification
- JWT Authentication with Refresh Token
- Multi-Factor Authentication (*Future*)

FR-002: User Management

The system shall allow **administrators** to manage the complete lifecycle of user accounts. This module is critical for onboarding new staff, managing role changes, and ensuring that only active, authorised users can access the system.

- Create new user accounts
- Edit existing user profiles
- Delete user accounts
- Assign and change roles
- Reset user passwords
- Activate and deactivate accounts

User Management is tightly coupled with FR-001, as account activation status directly affects authentication outcomes. Deactivated accounts must be immediately blocked from logging in.



The diagram above illustrates the complete user lifecycle managed across FR-001 and FR-002. Every user passes through these stages from initial account creation through to eventual deactivation or deletion, with authentication occurring at each active session.

Functional Requirements: Member & Membership Management

FR-003 FR-004

Modules **FR-003** and **FR-004** are central to GymX's core business logic. FR-003 governs how staff manage member records throughout their lifecycle, while FR-004 defines the membership plan structures that drive revenue. These modules are used daily by Receptionists, Admins, and Branch Managers.

FR-003: Member Management

The system shall allow staff to manage the complete **member lifecycle**, from initial registration through to archiving inactive accounts. This module supports photo uploads, profile management, and membership status controls including freezing and renewal.

- Register new members
- Update member profiles
- Upload profile photos
- View membership history
- Freeze memberships
- Renew memberships
- Archive inactive members

FR-004: Membership Management

The system shall support a **diverse range of membership plans** to cater to different customer segments. Each plan type may have different pricing, duration, and benefits. The system must allow Admins to configure and manage these plans flexibly.

- Monthly plans
- Quarterly plans
- Annual plans
- VIP plans
- Student plans
- Corporate memberships

Membership plans are linked to FR-009 (Payment Management) for billing and to FR-003 for member assignment.

Monthly

Standard short-term commitment

Quarterly

3-month plan with savings

Annual

Best value long-term plan

VIP

Premium all-access tier

Student

Discounted academic rate

Corporate

Business group packages

Functional Requirements: Attendance & Workout Management

FR-005

FR-006

FR-005 and **FR-006** address two critical operational areas: tracking member presence and delivering personalised fitness programming. Attendance management provides the data backbone for engagement analytics, while workout management empowers coaches to create and deliver structured training programmes.



FR-005: Attendance Management

The system shall provide a comprehensive **attendance tracking system** using QR codes and manual check-in methods. Duplicate scan detection prevents fraud, and automated reports give Branch Managers real-time visibility into member engagement patterns.

- Record check-in and check-out
- QR Code scanning support
- Generate attendance reports
- Detect duplicate scans



FR-006: Workout Management

The system shall allow **coaches** to create, assign, and manage personalised workout plans for members. Coaches can track progress over time, update routines as members advance, and archive completed plans for historical reference.

- Create workout plans
- Assign exercises to plans
- Track member progress
- Update and modify routines
- Archive completed plans

Attendance data from FR-005 feeds directly into the Reporting System (FR-014) and is visible on Branch Manager and Admin dashboards (FR-013). Workout plans from FR-006 are accessible to Members via their personal dashboard and to Coaches for ongoing progress tracking.

Functional Requirements: Nutrition, Classes & Payments

FR-007 FR-008 FR-009

These three modules address the **holistic member experience** – from nutrition planning and class booking to financial transactions. FR-007, FR-008, and FR-009 are among the most frequently used features by Members, Coaches, Nutritionists, and Accountants on a daily basis.



FR-007: Nutrition Management

Nutritionists can create personalised **meal plans**, calculate calorie requirements, track member weight, monitor BMI, and update nutrition plans as members progress. This module integrates with FR-006 to provide a complete fitness and nutrition programme.



FR-008: Class Management

The system shall support the full lifecycle of **fitness classes** – from creation and scheduling to coach assignment and online booking. Capacity limits prevent overbooking, and Members can book classes directly from their dashboard.



FR-009: Payment Management

The system shall handle all **financial transactions** including cash and online payments, invoice generation, transaction recording, refunds, and financial report exports. This module is critical for the Accountant role and integrates with FR-004 for membership billing.

Meal Plans FR-007	Class Booking FR-008	Online Payments FR-009
Invoices FR-009		Refunds FR-009

Functional Requirements: Infrastructure, Reporting & AI

- FR-010
- FR-011
- FR-012
- FR-013
- FR-014
- FR-015

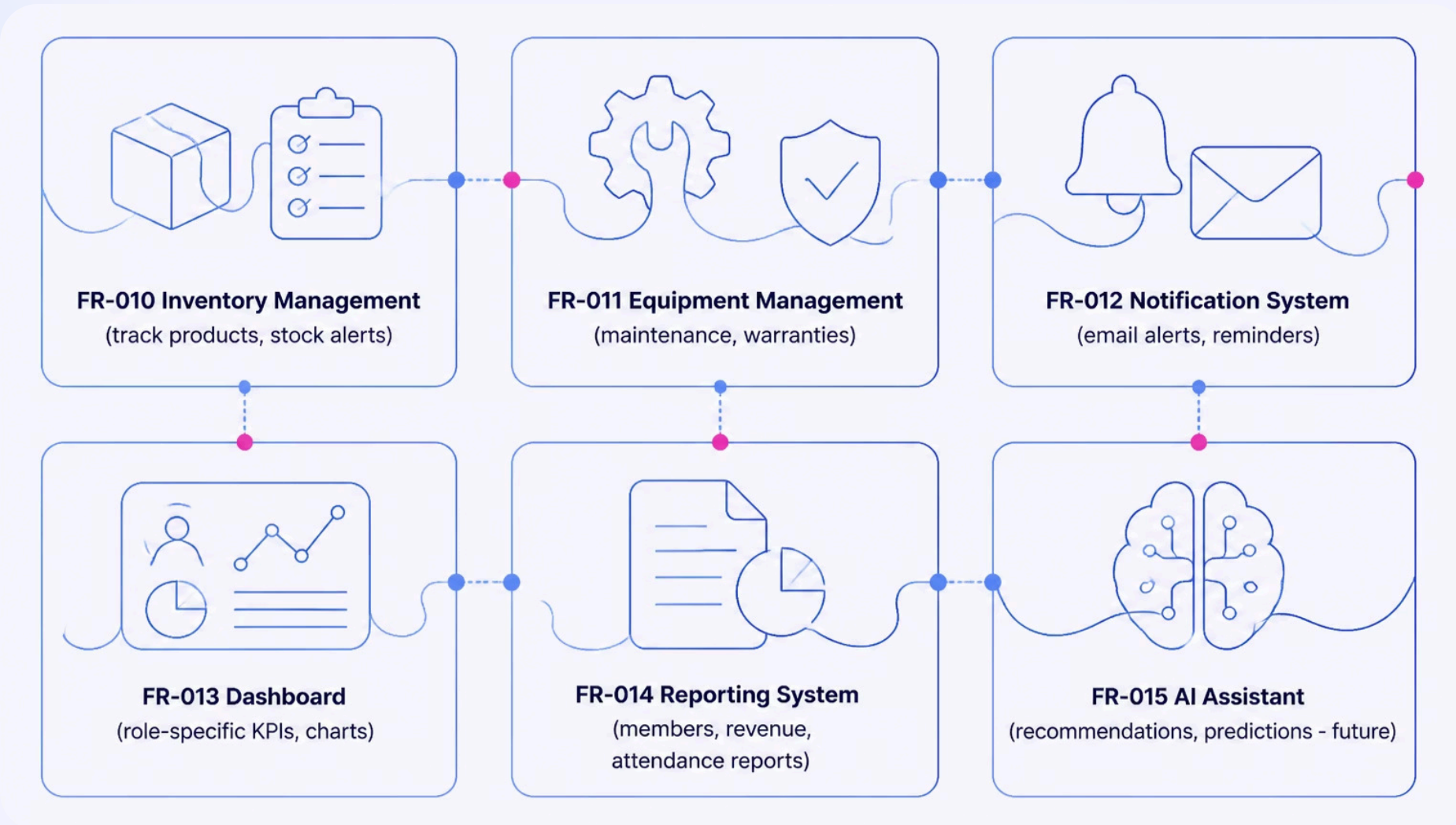
The final six functional modules cover the **infrastructure, intelligence, and reporting layers** of GymX. These modules ensure the platform is well-maintained, well-communicated, data-rich, and forward-looking. FR-015 represents the platform's future AI capabilities, marking GymX as a next-generation fitness management solution.

Infrastructure Modules

- FR-010 Inventory Management** — Track products, supplements, and equipment; monitor stock levels and generate stock alerts.
- FR-011 Equipment Management** — Register equipment, record maintenance history, track warranties, and schedule inspections.
- FR-012 Notification System** — Send email notifications for membership reminders, payment reminders, class reminders, and maintenance alerts.

Intelligence & Reporting Modules

- FR-013 Dashboard** — Provide role-specific dashboards with KPIs, charts, summaries, and quick actions for every user class.
- FR-014 Reporting System** — Generate comprehensive reports for members, attendance, revenue, memberships, coaches, classes, inventory, equipment, and branches.
- FR-015 AI Assistant** (*Future Enhancement*) — Recommend workout and nutrition plans, answer member questions, analyse performance, and predict membership renewals.



GymX is designed not just to manage a gym, but to **intelligently optimise every aspect** of its operations — from the moment a Guest registers to the moment an AI assistant predicts their next renewal. The 15 functional modules and 11 user classes together form a complete, scalable, and future-ready platform.